

Detection of Pathological Voice Disorders from Sustained Vowel Recordings

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Abstract—Objectives: This study aims to develop an automatic pathological voice disorder detection system using sustained vowel recordings and to evaluate the effectiveness of different machine learning and deep learning approaches for binary voice classification. Inspired by previous studies on cepstral-based pathological voice analysis, the proposed framework investigates whether acoustic features extracted from sustained phonation can distinguish healthy voices from different pathological conditions.

Methods: Voice recordings collected in WFDB format were preprocessed through signal normalization and feature extraction procedures. Acoustic representations were generated using Mel-Frequency Cepstral Coefficients (MFCCs) together with their first-order and second-order derivatives (Delta and Delta-Delta coefficients) to capture spectral and temporal characteristics of speech signals. The extracted features were standardized and used to train three classification models: Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN). A binary classification setting was adopted to distinguish healthy voices from pathological voices. Model performance was evaluated using 5-fold cross-validation and standard metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis.

Results: Experimental results showed that the Deep Neural Network achieved the highest average classification performance with an accuracy of 71.6% ($\pm 6.5\%$), followed by the Support Vector Machine with 70.6% ($\pm 4.3\%$), and the Gaussian Mixture Model with 64.4% ($\pm 7.1\%$). The findings indicate that cepstral-based acoustic features provide effective discriminative information for pathological voice detection, while temporal acoustic information further improves classification performance.

Conclusions: The study demonstrates that sustained vowel recordings combined with cepstral feature extraction can support automatic detection of pathological voice disorders. Among the evaluated approaches, DNN achieved the highest classification accuracy, while SVM provided competitive and stable performance for binary pathological voice detection. Future work may focus on incorporating pathology-specific acoustic descriptors, temporal deep learning architectures, and data augmentation techniques to further improve classification performance and clinical applicability.

I. INTRODUCTION

Human voice carries abundant physiological information regarding the condition of the vocal folds and the speech production system. Various pathological disorders, including vocal fold lesions, neurological diseases, and inflammatory conditions, may alter speech production by affecting vocal fold vibration, airflow, and resonance. These physiological changes are reflected in the acoustic characteristics of speech signals,

making voice analysis a promising non-invasive approach for assisting the diagnosis and monitoring of voice disorders.

Conventional clinical assessment of pathological voices primarily relies on perceptual evaluation performed by experienced clinicians together with laryngoscopic examination. Although these methods provide reliable diagnostic information, they require specialized equipment, clinical expertise, and considerable examination time. Consequently, automatic pathological voice detection based on speech signal analysis has attracted increasing attention as an objective, low-cost, and easily deployable screening tool that may assist clinicians during preliminary assessment.

Recent advances in speech signal processing and machine learning have enabled significant improvements in automatic voice disorder detection systems. Numerous studies have demonstrated that acoustic features extracted directly from sustained vowel recordings contain valuable information for distinguishing healthy voices from pathological voices. Among various acoustic representations, Mel-Frequency Cepstral Coefficients (MFCCs) remain one of the most widely adopted features because they effectively characterize the perceptually relevant spectral envelope of speech while maintaining relatively low computational complexity [1].

Several machine learning approaches have been investigated for pathological voice detection. Support Vector Machine (SVM) has been extensively employed because of its strong generalization capability under limited training data, whereas Gaussian Mixture Model (GMM) provides probabilistic modeling of acoustic feature distributions and has been widely applied in speech and speaker recognition tasks [2]. More recently, deep learning techniques have demonstrated the ability to learn complex nonlinear relationships between acoustic features and pathological conditions. For example, Fang *et al.* [3] proposed a Deep Neural Network (DNN) using cepstral features and reported improved performance compared with several conventional classifiers.

Despite these advances, several challenges remain. Existing studies employ different datasets, feature extraction strategies, classifier architectures, and evaluation protocols, making direct comparison difficult. Furthermore, although temporal cepstral information derived from Delta and Delta-Delta coefficients has been widely adopted in speech processing, its contribution to pathological voice detection has not been systematically investigated under identical experimental conditions across both conventional machine learning and deep learning models. Consequently, the relative effectiveness of different MFCC-

based feature representations for pathological voice detection remains insufficiently understood.

Motivated by these observations, this study develops an automatic pathological voice detection framework based on sustained vowel recordings and performs a systematic comparison of three MFCC-based acoustic feature configurations, namely MFCC, MFCC combined with Delta coefficients, and MFCC combined with both Delta and Delta-Delta coefficients. These feature representations are evaluated using three representative classifiers, including Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN), under the same preprocessing pipeline and stratified 5-fold cross-validation protocol.

The main contributions of this study are summarized as follows:

- A complete automatic pathological voice detection framework is developed using sustained vowel recordings, including audio preprocessing, MFCC-based feature extraction, feature standardization, and binary classification.
- Three MFCC-based acoustic feature configurations (MFCC, MFCC + Delta, and MFCC + Delta + Delta-Delta) are systematically compared under identical experimental conditions to investigate the contribution of temporal cepstral information.
- Conventional machine learning approaches (SVM and GMM) and a Deep Neural Network (DNN) are comparatively evaluated using the same dataset and validation protocol, enabling a consistent assessment of their relative performance.
- Experimental results provide practical insights into the effectiveness of different MFCC-based feature representations for automatic pathological voice detection and establish a reproducible baseline for future investigations.

II. METHODS

A. Study Subjects

The experiments were conducted using the Voice-ICAR-Federico II Database, which contains sustained vowel recordings collected from both healthy subjects and patients diagnosed with pathological voice disorders. Each recording is provided in WaveForm DataBase (WFDB) format together with metadata describing demographic information and diagnostic labels.

Sustained vowel phonation was selected because it minimizes linguistic variability while preserving stable vocal fold vibration patterns. Consequently, acoustic variations observed in the recordings are more likely to reflect physiological abnormalities rather than differences in speech content.

Incomplete recordings or samples without valid diagnostic labels were excluded during data preparation. The remaining recordings were divided into two categories:

To reduce sampling bias, all experiments were evaluated using stratified 5-fold cross-validation, ensuring that both classes maintained similar proportions in every fold.

Table I
SUMMARY OF THE PATHOLOGICAL VOICE DATASET

Characteristic	Value
Dataset	MEEI Voice Disorders Database
Recording type	Sustained vowel /a/
Sampling frequency	16 kHz
Healthy recordings	57
Pathological recordings	151
Total recordings	208
Classification task	Binary classification

B. Audio Preprocessing

Prior to feature extraction, several preprocessing operations were performed to improve signal consistency and reduce unwanted variability.

Each WFDB recording was first converted into a one-dimensional speech signal. Since several recordings contain multiple channels, only the first channel was retained for analysis.

All recordings were converted to a unified sampling frequency before feature extraction. Each speech signal was normalized to reduce amplitude variations across speakers. Silence regions at the beginning and end of each recording were removed to retain only the voiced segments. These preprocessing procedures reduce recording variability and improve the consistency of subsequent acoustic feature extraction.

Amplitude normalization was then applied to reduce recording-level intensity differences among speakers. All recordings were subsequently resampled to a fixed sampling frequency in order to obtain identical spectral resolution.

Finally, recordings were standardized to equal duration through truncation or zero-padding. Standardizing signal length enables fixed-dimensional feature extraction and facilitates subsequent machine learning training.

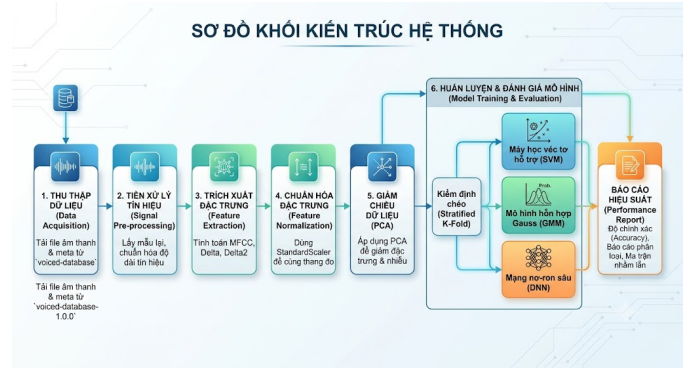


Figure 1. Block diagram of the proposed automatic pathological voice detection system including preprocessing, feature extraction, and model classification.

C. Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) were employed as the primary acoustic representation because they effectively characterize the spectral envelope of speech and

have demonstrated excellent performance in pathological voice analysis.

Three different feature configurations were investigated:

- MFCC
- MFCC + Delta
- MFCC + Delta + Delta-Delta

For each recording, 13 MFCC coefficients were extracted from short-time speech frames. Delta coefficients were incorporated to describe first-order temporal variations, whereas Delta-Delta coefficients represent second-order temporal dynamics.

To generate fixed-length feature vectors, the mean and standard deviation of each coefficient sequence were computed over the entire recording.

Detailed mathematical definitions of MFCC, Delta coefficients, and Delta-Delta coefficients are provided in Appendix B.

MFCC features were extracted using the Librosa library. The speech signal was segmented into 32-ms frames with a 10-ms frame shift using a Hanning window. A 512-point Fast Fourier Transform (FFT) was applied, followed by a Mel filter bank. The first 13 MFCC coefficients were retained. First-order (Delta) and second-order (Delta-Delta) coefficients were subsequently computed to capture temporal dynamics.

D. Classification Models

Three representative classification models were evaluated in this study.

1) *Support Vector Machine*: Support Vector Machine (SVM) is a supervised learning algorithm that determines an optimal separating hyperplane by maximizing the margin between different classes.

Because of its strong generalization capability and robustness on relatively small datasets, SVM has been widely adopted in speech processing and medical classification problems.

2) *Gaussian Mixture Model*: Gaussian Mixture Model (GMM) is a probabilistic classifier that models feature distributions as weighted combinations of Gaussian probability density functions.

Unlike discriminative classifiers, GMM estimates the underlying statistical distribution of acoustic features, making it a widely used baseline method in speech and speaker recognition research.

3) *Deep Neural Network*: Deep Neural Network (DNN) was employed to learn nonlinear mappings between acoustic feature vectors and voice categories.

The proposed network consists of three fully connected hidden layers containing 256, 128, and 64 neurons, respectively. Rectified Linear Unit (ReLU) activation was adopted in all hidden layers, while the output layer utilized the Softmax function to estimate posterior probabilities of the two voice classes.

Model parameters were optimized using the Adam optimizer together with sparse categorical cross-entropy loss.

Detailed mathematical formulations of the DNN architecture are presented in Appendix C.

E. Experimental Setup

Before model training, all feature vectors were standardized using z-score normalization to obtain zero mean and unit variance.

Binary classification experiments were performed using stratified 5-fold cross-validation. During each iteration, four folds were used for training and one fold was reserved for testing. The reported performance corresponds to the average results obtained across all folds.

Three classifiers, namely SVM, GMM, and DNN, were evaluated under identical experimental conditions for fair comparison.

Classification performance was assessed using the following evaluation metrics:

- Accuracy
- Precision
- Recall
- F1-score

These metrics provide complementary information regarding overall classification performance as well as the capability of correctly identifying pathological voice recordings.

III. RESULTS AND DISCUSSION

This section presents the experimental results obtained from the three evaluated classifiers, namely Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN). The influence of different acoustic feature configurations is first analyzed, followed by a comparison of classification models and a discussion of the obtained findings.

A. Performance under Different Feature Configurations

To investigate the contribution of temporal cepstral information to pathological voice detection, three MFCC-based feature configurations were evaluated: (i) static MFCCs, (ii) MFCC combined with first-order Delta coefficients, and (iii) MFCC combined with both Delta and Delta-Delta coefficients. The average classification accuracy obtained using stratified 5-fold cross-validation is presented in Table II.

Table II
AVERAGE CLASSIFICATION ACCURACY UNDER DIFFERENT FEATURE CONFIGURATIONS

Feature Configuration	SVM	GMM	DNN
MFCC	0.653 $\pm$ 0.055	0.649 $\pm$ 0.088	0.615 $\pm$ 0.054
MFCC + Delta	0.706 $\pm$ 0.043	0.644 $\pm$ 0.071	0.716 $\pm$ 0.065
MFCC + Delta + Delta-Delta	0.702 $\pm$ 0.026	0.606 $\pm$ 0.080	0.682 $\pm$ 0.043

The experimental results indicate that incorporating first-order temporal cepstral information generally improves classification performance compared with using static MFCC features alone under the adopted evaluation protocol. In particular, the inclusion of Delta coefficients increased the average classification accuracy of both SVM and DNN, suggesting that

temporal spectral variations provide complementary information beyond static cepstral representations.

Among the three evaluated feature configurations, MFCC combined with Delta coefficients consistently achieved the highest average accuracy for the discriminative classifiers. Specifically, the DNN obtained the highest overall performance with an average accuracy of 71.6%, followed closely by the SVM with 70.6%. These findings suggest that first-order temporal dynamics capture additional characteristics associated with pathological voice production, thereby improving the discriminative capability of MFCC-based acoustic features.

In contrast, incorporating Delta-Delta coefficients did not consistently improve classification performance. Although second-order temporal derivatives provide additional dynamic information, both the GMM and DNN exhibited reduced accuracy when these features were included. One possible explanation is that the increased feature dimensionality introduces redundant or less informative descriptors that are difficult to exploit effectively under the available training data. Consequently, the additional complexity introduced by Delta-Delta coefficients may outweigh their potential benefits for the evaluated dataset.

To provide a more comprehensive evaluation, the detailed classification metrics of the best-performing configuration (DNN with MFCC + Delta) are summarized in Table III.

Table III
DETAILED PERFORMANCE METRICS OF THE OPTIMAL CLASSIFIER (DNN WITH MFCC + DELTA)

Class / Metric	Precision	Recall	F1-Score	Support
Healthy	0.374	0.232	0.282	57
Pathological	0.744	0.848	0.792	151
Weighted Avg	0.643	0.678	0.652	208
Overall Accuracy	0.678 $\pm$ 0.061			

The class-wise evaluation reveals an important characteristic of the dataset, namely a substantial class imbalance, with pathological recordings accounting for approximately 72.6% of the available samples (151 pathological recordings versus 57 healthy recordings). This imbalance influences the classifier’s predictive behavior and should therefore be considered when interpreting the experimental results.

As shown in Table III, the proposed DNN achieved a Recall (Sensitivity) of 0.848 and an F1-score of 0.792 for the pathological class. From a clinical screening perspective, higher sensitivity is desirable because it reduces the likelihood of failing to identify individuals with potential voice disorders during preliminary assessment. These results indicate that the proposed framework is effective in detecting pathological voice samples within the evaluated dataset.

Conversely, the recall for the healthy class remained relatively low (0.232), indicating that healthy voice recordings were more difficult to classify correctly. A plausible explanation is the limited number of healthy samples available for training, which may have restricted the model’s ability to learn the full variability of normal phonation. In addition,

the class imbalance may have biased the optimization process toward the majority pathological class. These observations suggest that future work could benefit from incorporating class-balancing strategies, such as synthetic oversampling, cost-sensitive learning, or additional healthy voice recordings, to improve classification performance for the minority class while maintaining high pathological sensitivity.

B. Comparison of Classification Models

To further investigate the effectiveness of different classification algorithms, the three evaluated models were compared using the optimal acoustic feature configuration identified in the previous experiment. The comparison focuses on the ability of each classifier to exploit MFCC-based representations for distinguishing healthy and pathological voice recordings.

Among the evaluated classifiers, the Deep Neural Network (DNN) achieved the highest average classification accuracy of 71.6% when trained using the MFCC + Delta feature configuration. This result suggests that nonlinear representation learning is advantageous for modeling the complex acoustic characteristics associated with pathological voice production. By learning hierarchical feature representations, the DNN is able to capture relationships that may not be adequately represented by conventional linear or probabilistic classifiers.

The Support Vector Machine (SVM) produced a comparable average accuracy of 70.6%, differing by only one percentage point from the DNN. Although the overall classification performance was slightly lower, the SVM exhibited a smaller standard deviation across the five cross-validation folds, indicating relatively stable performance under different training and testing partitions. This observation is consistent with the well-established robustness of SVMs when applied to medium-sized datasets and high-dimensional acoustic feature spaces.

The comparative analysis also highlights the influence of classifier complexity on pathological voice detection. While deep learning models possess greater representational capacity, their performance depends on the availability of sufficient training data and appropriate feature representations. Conversely, SVM provides competitive performance with considerably lower model complexity and computational requirements, making it an attractive alternative for applications where computational resources or training data are limited.

Overall, the experimental results demonstrate that both discriminative classifiers achieved higher classification performance than the probabilistic GMM model for the evaluated dataset. Although the DNN obtained the highest average accuracy, the relatively small performance difference between the DNN and SVM suggests that both approaches are suitable candidates for MFCC-based pathological voice detection. Therefore, the choice between these classifiers may ultimately depend on practical considerations such as computational cost, model interpretability, and deployment requirements rather than classification accuracy alone.

C. Comparison with Related Works

To better position the proposed framework within the existing literature, its performance was compared with several

representative studies on pathological voice detection using MFCC-based acoustic features. Table IV summarizes the reported classification accuracies together with the corresponding feature extraction methods and classifiers.

Table IV

COMPARISON WITH REPRESENTATIVE STUDIES ON PATHOLOGICAL VOICE DETECTION

Study	Features	Classifier	Accuracy (%)
Lee et al. [1]	MFCC	GMM	64.9
Harar et al. [2]	Acoustic features	SVM	76.5
Fang et al. [3]	MFCC + Delta	DNN	68.2
Proposed	MFCC + Delta	DNN	71.6

Direct numerical comparisons with previous studies should be interpreted cautiously because different datasets, recording conditions, preprocessing procedures, feature extraction methods, and evaluation protocols were employed. Consequently, the reported accuracies provide a general reference rather than a definitive ranking of different approaches.

Compared with the DNN-based approach proposed by Fang *et al.* [3], the proposed framework achieved a higher average classification accuracy (71.6% versus 68.2%). Although both studies employed MFCC-derived acoustic features together with deep neural networks, the present results suggest that incorporating first-order temporal cepstral information may provide complementary discriminative information beyond static MFCC representations under the evaluated experimental setting. Nevertheless, differences in datasets and experimental protocols should be considered when interpreting this comparison.

The proposed GMM classifier achieved an average accuracy comparable to that reported by Lee *et al.* [1]. While this consistency provides additional confidence in the implementation of the probabilistic baseline, the overall performance remained lower than that of the evaluated discriminative classifiers. This observation may reflect the limited capability of probabilistic density modeling to represent the complex nonlinear distributions of pathological voice features within the adopted dataset.

Harar *et al.* [2] reported a higher classification accuracy using an SVM classifier combined with a richer set of handcrafted acoustic descriptors, including perturbation, spectral, and noise-related features. In contrast, the present study intentionally focuses on MFCC-based cepstral representations to systematically evaluate the contribution of temporal cepstral information while maintaining a relatively simple and reproducible feature extraction pipeline. The performance difference between the two studies may therefore be partially explained by the additional complementary acoustic information incorporated in the handcrafted feature set.

Overall, the comparative analysis indicates that the proposed framework achieves competitive performance relative to several representative MFCC-based pathological voice detection systems reported in the literature. More importantly, the experimental results provide a consistent comparison of multiple classifiers under identical preprocessing, feature extraction,

and evaluation protocols. Such a controlled experimental setting facilitates a clearer assessment of the contribution of different MFCC-based feature configurations while establishing a reproducible baseline for future investigations.

D. Discussion

The experimental results demonstrate that the incorporation of temporal cepstral information can improve the effectiveness of MFCC-based pathological voice detection. Among the evaluated feature configurations, the combination of MFCC and first-order Delta coefficients consistently produced the highest classification performance for both the SVM and DNN classifiers. These findings suggest that temporal variations in cepstral features provide complementary information beyond static spectral representations, thereby improving the discriminative capability of acoustic features for distinguishing healthy and pathological voices.

In contrast, incorporating second-order Delta-Delta coefficients did not lead to further performance improvements. One possible explanation is that the additional temporal descriptors increase feature dimensionality without introducing sufficient discriminative information for the evaluated dataset. Under limited training data, the higher-dimensional feature space may also increase model complexity and reduce generalization performance. These observations indicate that richer feature representations do not necessarily translate into improved classification accuracy, particularly when the available data are relatively limited.

The comparative evaluation further demonstrates that discriminative learning approaches are better suited to the adopted feature representations than probabilistic density modeling. The DNN achieved the highest average classification accuracy, while the SVM produced comparable performance with lower computational complexity and relatively stable results across the cross-validation folds. These findings suggest that both classifiers are appropriate choices for MFCC-based pathological voice detection, with the final selection depending on application-specific requirements such as computational resources, model interpretability, and deployment constraints.

From a clinical perspective, the proposed framework achieved relatively high sensitivity for detecting pathological voices, indicating its potential usefulness as a computer-aided screening tool. Early identification of voice disorders may facilitate timely referral for further clinical examination, thereby assisting healthcare professionals during preliminary assessment. Nevertheless, the relatively low recall observed for healthy voices indicates that additional efforts are required to improve the balance between sensitivity and specificity before practical clinical deployment.

Several limitations of the present study should also be acknowledged. First, the experiments were conducted using a single publicly available dataset containing sustained vowel recordings, which may not fully represent the diversity of real-world clinical speech. Second, the dataset exhibits a noticeable class imbalance that may influence classifier learning and prediction. Third, only MFCC-based acoustic features

were investigated, whereas complementary descriptors, such as perturbation, spectral, prosodic, or self-supervised speech representations, may further improve classification performance. Finally, no statistical significance analysis was performed to determine whether the observed performance differences among classifiers were statistically significant. Future work should address these limitations by evaluating larger and more diverse datasets, incorporating additional acoustic representations, applying statistical hypothesis testing, and investigating recent deep learning architectures, including convolutional, recurrent, and transformer-based models.

E. Limitations

Several limitations should be considered when interpreting the experimental results.

First, the dataset size remains relatively limited for deep learning applications. Larger datasets would enable more reliable optimization and improve model generalization.

Second, only sustained vowel recordings were analyzed. Continuous speech may contain additional prosodic and phonatory information that could further improve pathological voice detection.

Third, this study focused primarily on MFCC-based acoustic features. Other clinically relevant descriptors, including jitter, shimmer, harmonic-to-noise ratio (HNR), and glottal parameters, were not incorporated.

Finally, only three classification algorithms were investigated. More advanced deep learning architectures, such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, Transformer-based models, or self-supervised speech representations, may provide further improvements in classification performance.

Overall, the proposed framework demonstrates that MFCC-based acoustic representations combined with machine learning and deep learning techniques constitute an effective baseline for automatic pathological voice detection. Future work will investigate richer acoustic representations, larger datasets, and more sophisticated neural architectures to further improve detection accuracy and clinical applicability.

IV. CONCLUSION

This study presented an automatic pathological voice detection framework based on MFCC-derived acoustic features and three representative classification algorithms, namely Support Vector Machine (SVM), Gaussian Mixture Model (GMM), and Deep Neural Network (DNN). A systematic comparison of three feature configurations, including MFCC, MFCC with Delta coefficients, and MFCC with both Delta and Delta-Delta coefficients, was conducted under a unified preprocessing pipeline and stratified five-fold cross-validation protocol.

The experimental results indicate that incorporating first-order temporal cepstral information generally improves classification performance compared with using static MFCC features alone. Among the evaluated models, the DNN combined with MFCC and Delta coefficients achieved the highest average classification accuracy of 71.6%, while the SVM

obtained comparable performance with lower computational complexity. In contrast, the inclusion of Delta-Delta coefficients did not consistently improve classification accuracy, suggesting that additional temporal descriptors may increase feature dimensionality without providing sufficient complementary information for the evaluated dataset.

Overall, the findings suggest that first-order temporal cepstral dynamics provide useful complementary information for automatic pathological voice detection and that discriminative learning approaches are well suited to exploiting these acoustic representations. More importantly, the study provides a consistent comparison of multiple classifiers under identical preprocessing, feature extraction, and evaluation protocols, establishing a reproducible baseline for future investigations.

Several limitations should be acknowledged. The proposed framework was evaluated using a single publicly available dataset and focused exclusively on MFCC-based acoustic features extracted from sustained vowel recordings. In addition, the class imbalance of the dataset and the absence of statistical significance analysis may influence the interpretation of the reported performance differences. Future research should investigate larger and more diverse clinical datasets, integrate complementary acoustic and self-supervised speech representations, perform statistical validation of experimental results, and evaluate more advanced deep learning architectures to further improve the robustness and generalizability of automatic pathological voice detection systems.

APPENDIX A SIGNAL VISUALIZATION

This appendix presents visual comparisons between healthy and pathological voice recordings.

Figure 2 illustrates the waveform of sustained vowel recordings. Healthy voices generally exhibit relatively stable amplitude patterns, whereas pathological voices show larger fluctuations caused by irregular vocal fold vibration.

Figure 3 presents the corresponding spectrograms. Compared with healthy voices, pathological speech demonstrates less regular harmonic structures together with greater spectral dispersion, indicating abnormal phonatory behavior.

APPENDIX B MFCC FEATURE FORMULATION

Mel-Frequency Cepstral Coefficients (MFCCs) are extracted from short-time speech frames to represent perceptually meaningful spectral information.

The Mel frequency scale is defined as

$$m(f) = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$

where f denotes the physical frequency in Hertz.

The first-order temporal derivative (Delta coefficient) is computed as

$$\Delta c_t = \frac{\sum_{n=1}^N n(c_{t+n} - c_{t-n})}{2 \sum_{n=1}^N n^2}$$

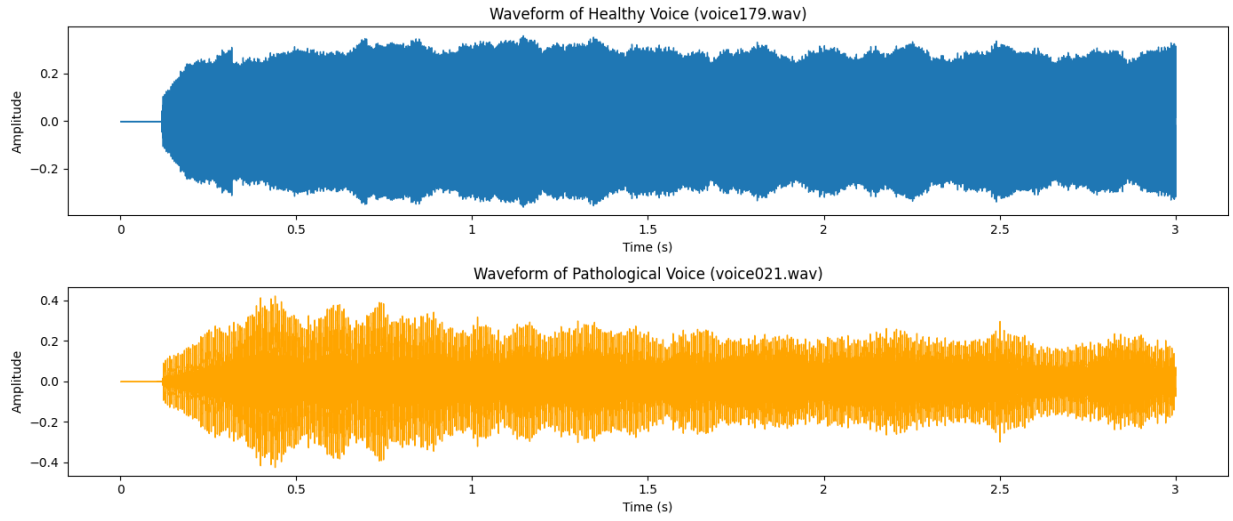


Figure 2. Waveform Comparison (Healthy vs. Pathological Voice)

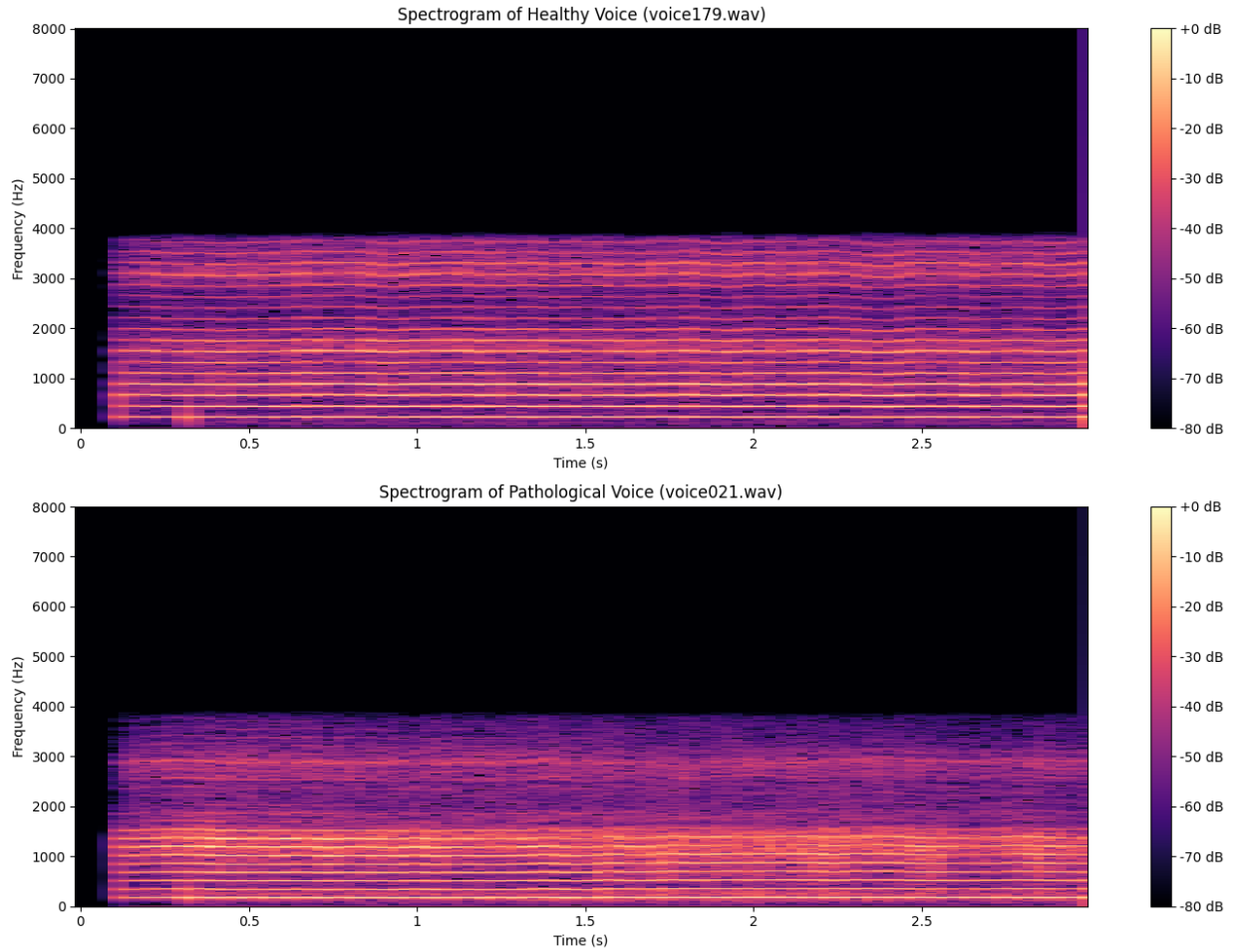


Figure 3. Spectrogram Comparison (Healthy vs. Pathological Voice)

The second-order derivative (Delta-Delta coefficient) is obtained by applying the same operation to the Delta coefficients.

For each recording, statistical aggregation is performed to

generate fixed-length feature vectors:

$$\mathbf{Feature} = [\mu(\text{MFCC}), \sigma(\text{MFCC}), \mu(\Delta), \sigma(\Delta), \mu(\Delta^2), \sigma(\Delta^2)]$$

where μ and σ denote the mean and standard deviation, respectively.

APPENDIX C DEEP NEURAL NETWORK FORMULATION

The Deep Neural Network employed in this study consists of multiple fully connected hidden layers.

The first hidden layer is computed as

$$h_1 = f(W_1x + b_1)$$

where W_1 represents the weight matrix, b_1 denotes the bias vector, and $f(\cdot)$ is the activation function.

Subsequent hidden layers are computed recursively as

$$h_{l+1} = f(W_lh_l + b_l)$$

The output layer applies the Softmax function

$$\hat{y}_j = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}}$$

where $K = 2$ represents the two output classes (healthy and pathological).

The network parameters are optimized by minimizing the sparse categorical cross-entropy loss

$$L = - \sum_{i=1}^N \sum_{j=1}^K y_{ij} \log(\hat{y}_{ij})$$

using the Adam optimization algorithm.

APPENDIX D EXPERIMENTAL CONFIGURATION

Table V summarizes the principal hyperparameters used during model training.

Table V
EXPERIMENTAL CONFIGURATION

Parameter	Value
Cross-validation	Stratified 5-fold
Feature normalization	Z-score
MFCC coefficients	13
Feature sets	MFCC, MFCC+Delta, MFCC+Delta+Delta ²
Optimizer (DNN)	Adam
Loss function	Sparse Categorical Cross-Entropy
Hidden layers	256–128–64
Activation	ReLU
Output activation	Softmax
Classification	Binary

Additional evaluation metrics including precision, recall, F1-score, confusion matrices, and classification reports can also be included in this appendix for comprehensive performance analysis.

REFERENCES

- [1] J. Lee, S. Choi, and H. Kim, "Pathological voice detection using mel-frequency cepstral coefficients and machine learning algorithms," *Applied Sciences*, vol. 10, no. 19, p. 7051, 2020.
- [2] P. Harar, J. B. Alonso-Hernandez, T. Bocklet, J. R. Orozco-Arroyave, and E. Nöth, "Automatic detection of voice disorders using modulation spectrum-optimized metrics," *IEEE Journal of Biomedical and Health Informatics*, vol. 23, no. 4, pp. 1607–1615, 2019.
- [3] S.-H. Fang, Y. Tsao, M.-J. Hsiao, J.-Y. Chen, Y.-H. Lai, F.-C. Lin, and C.-T. Wang, "Detection of pathological voice using cepstrum vectors: A deep learning approach," *Journal of Voice*, vol. 33, no. 5, pp. 634–641, 2019.